

GDP and life expectancy globally between countries over the years

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Part 1 - Identify a Social Problem

1.1 Describe the Social Problem

Differences in life expectancy between countries are one of the clearest forms of global inequality. A child born in a high-income country can expect to live around fifteen to twenty years longer than a child born in a low-income country, and this gap has persisted for decades. The classic explanation links population health to national income: (Preston, 1975) showed that life expectancy rises steeply with GDP per capita at low

income levels and flattens among richer countries, which implies that economic growth buys the most health where countries are poorest (Deaton, 2003)]. A key lever for policy is therefore knowing how tightly health follows income, because that determines whether health gaps should be fought with economic development, with health spending, or both.

What is less well established is how this relationship behaves during a global shock. The COVID-19 pandemic caused the first worldwide drop in life expectancy in decades, and the losses were very unequal between countries (Schöley, 2022), yet income alone did not fully predict who lost the most. In this report we quantify the relationship between GDP per capita and life expectancy for ten countries across all income levels between 2019 and 2023, and we examine how life expectancy changed around the 2020 outbreak of the COVID-19 pandemic.

Part 2 - Data Sourcing

2.1 Load in the data

We load both indicators directly from the World Bank API using the WDI package, so no raw data files need to be stored in this repository.

2.2 Provide a short summary of the dataset(s)

This analysis uses two datasets from the World Bank's World Development Indicators. Both were downloaded as CSV files, and each can also be identified by its WDI indicator code.

The first dataset reports GDP per capita, PPP, in constant international dollars (indicator NY.GDP.PCAP.PP.KD). Measuring income in purchasing power parity terms and holding prices constant means GDP per capita can be compared fairly between countries and across years, without being thrown off by exchange rate swings or inflation.

The second dataset reports life expectancy at birth, total, in years (indicator SP.DYN.LE00.IN). This is the average number of years a newborn would live if the mortality patterns of that year stayed the same throughout its life.

In their raw form, each file has one row per economy (around 260 countries and regional aggregates) and one column per year going back to 1960, plus identifier columns for country name, country code, and indicator name and code. For this report we narrowed the data down to ten countries that cover a range of income levels: the United States, the Netherlands, Germany, Japan, China, India, Brazil, South Africa, Nigeria, and Australia. We also limited it to the five years from 2019 to 2023, which is the window around the COVID-19 outbreak. After reshaping both files into long format and merging them on country and year, the final working dataset (combined_data) has 50 observations (10 countries across 5 years) and five columns: country, country_code, year, gdp_per_capita in constant international dollars, and life_expectancy in years.

Part 3 - Our findings

3.1 Describe the variables

After cleaning and merging the datasets, the final dataset contains information on 10 selected countries from 2019 to 2023. The main variables are country, year, GDP per capita, and life expectancy. In addition to these original variables, we created two new categorical variables to support our analysis. The variable development_group separates countries into developed and developing countries, which is used for the sub-population comparison. The variable covid_period divides the years into pre-COVID, COVID period, and post-COVID recovery, which is used to analyse the effect of the COVID-19 period on life expectancy.

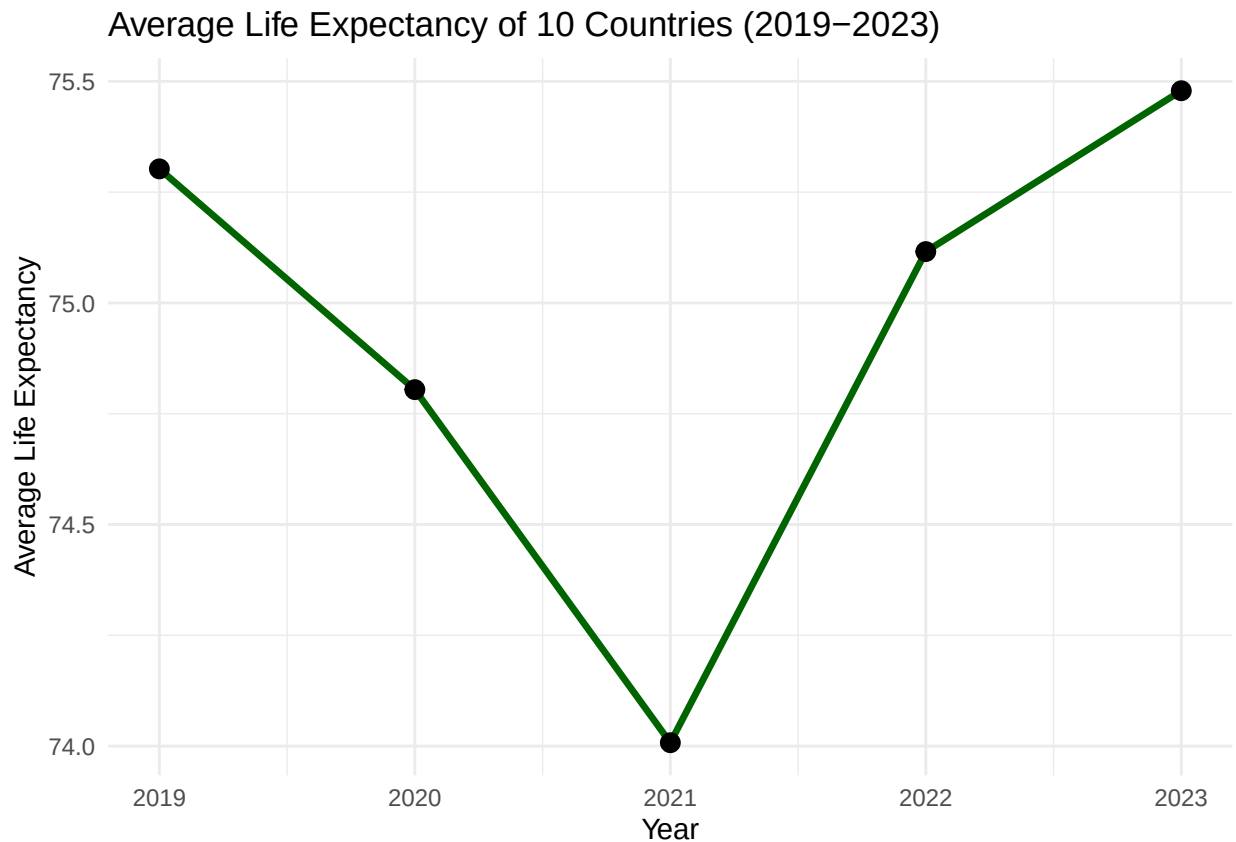
3.2 Generate necessary variables

Variable 1 To add analytical value to the dataset, we created two new categorical variables. First, we constructed `development_group`, which classifies the ten selected countries into developed and developing countries. This variable was used in the sub-population analysis to compare GDP per capita and life expectancy across different development groups.

Variable 2 Second, we created `covid_period`, which classifies the years into pre-COVID, COVID period, and post-COVID recovery. This variable was used to analyse how average life expectancy changed before, during, and after the COVID-19 pandemic.

3.3 Visualize temporal variation

Figure X shows the average life expectancy of the selected 10 countries between 2019 and 2023. Average life expectancy declined from 2019 to 2021, reaching its lowest point in 2021 at around 74.0 years. After 2021, it increased again and reached the highest level in 2023, at about 75.5 years. This pattern suggests a temporary decline followed by recovery. One possible explanation is that the decline around 2020–2021 may be related to global health disruptions, but further evidence would be needed to confirm this.



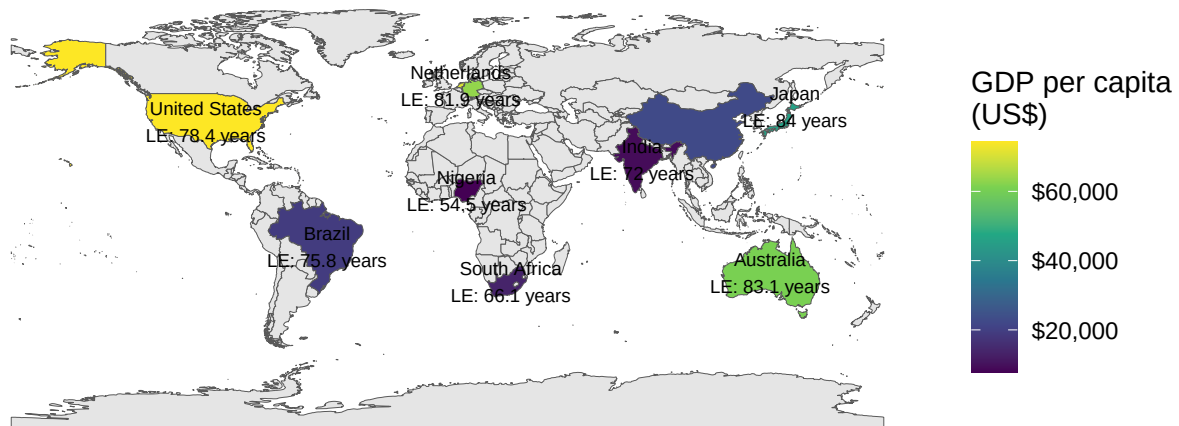
3.4 Visualize spatial variation

Figure X visualises the spatial variation in GDP per capita across the 10 selected countries in 2023. The map shows that GDP per capita is not evenly distributed across regions. Countries such as the United States, Australia, Sweden, and Japan have relatively high GDP per capita, while countries such as Nigeria, India, and South Africa show lower values. The life expectancy labels also suggest a similar regional pattern: countries

with higher GDP per capita generally tend to have longer life expectancy. However, some differences remain between countries, indicating that economic development is not the only factor affecting health outcomes.

GDP per Capita Across 10 Selected Countries (2023)

Map colours indicate GDP per capita; labels show life expectancy



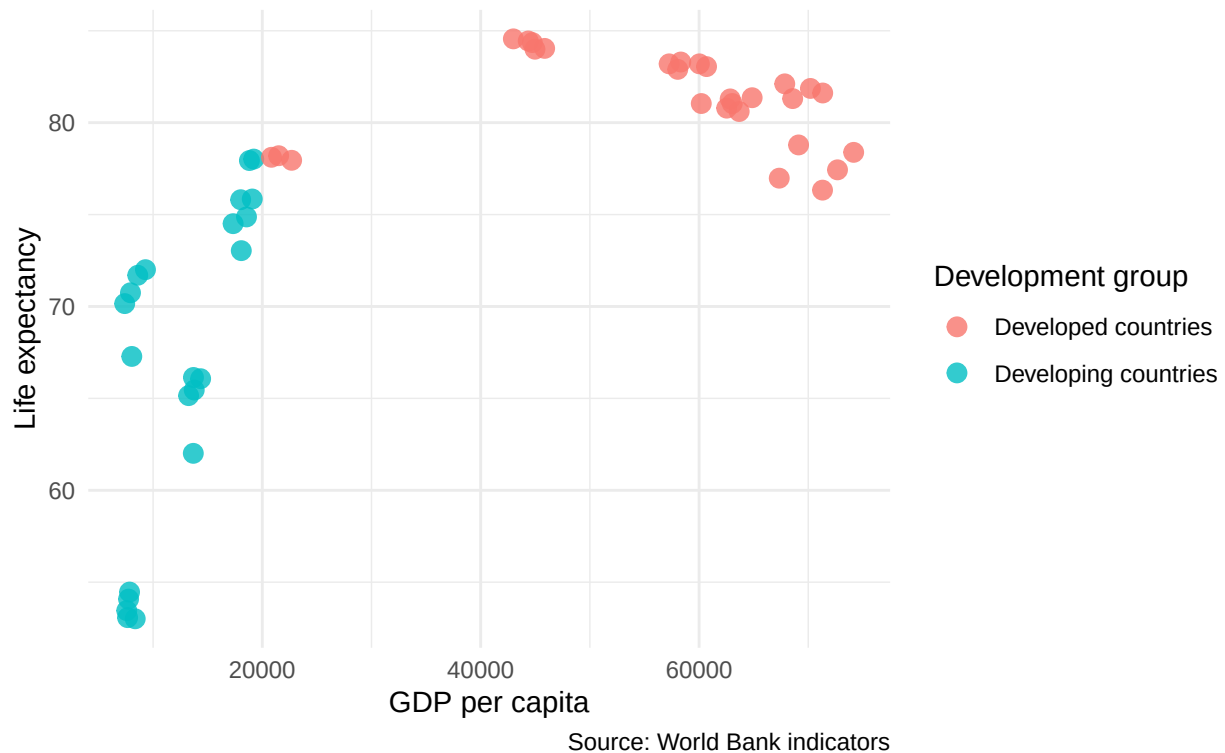
Source: cleaned and merged dataset. Authors' own calculations.

3.5 Visualize sub-population variation

The graph highlights a clear difference between developed and developing countries. Developed countries generally show both higher GDP per capita and higher life expectancy, whereas developing countries tend to have lower values for both indicators. This pattern suggests that higher levels of economic development are associated with better health outcomes and longer lives.

```
## Rows: 50
## Columns: 5
## $ country      <chr> "Australia", "Australia", "Australia", "Australia", "A~
## $ country_code <chr> "AUS", "AUS", "AUS", "AUS", "AUS", "BRA", "BRA", "BRA"~
## $ year         <dbl> 2019, 2020, 2021, 2022, 2023, 2019, 2020, 2021, 2022, ~
## $ gdp_per_capita <dbl> 58048.619, 57260.465, 58327.415, 60029.133, 60684.960,~
## $ life_expectancy <dbl> 82.90000, 83.20000, 83.30000, 83.20000, 83.05122, 75.8~
```

GDP per Capita and Life Expectancy by Development Group Comparison between developed and developing countries

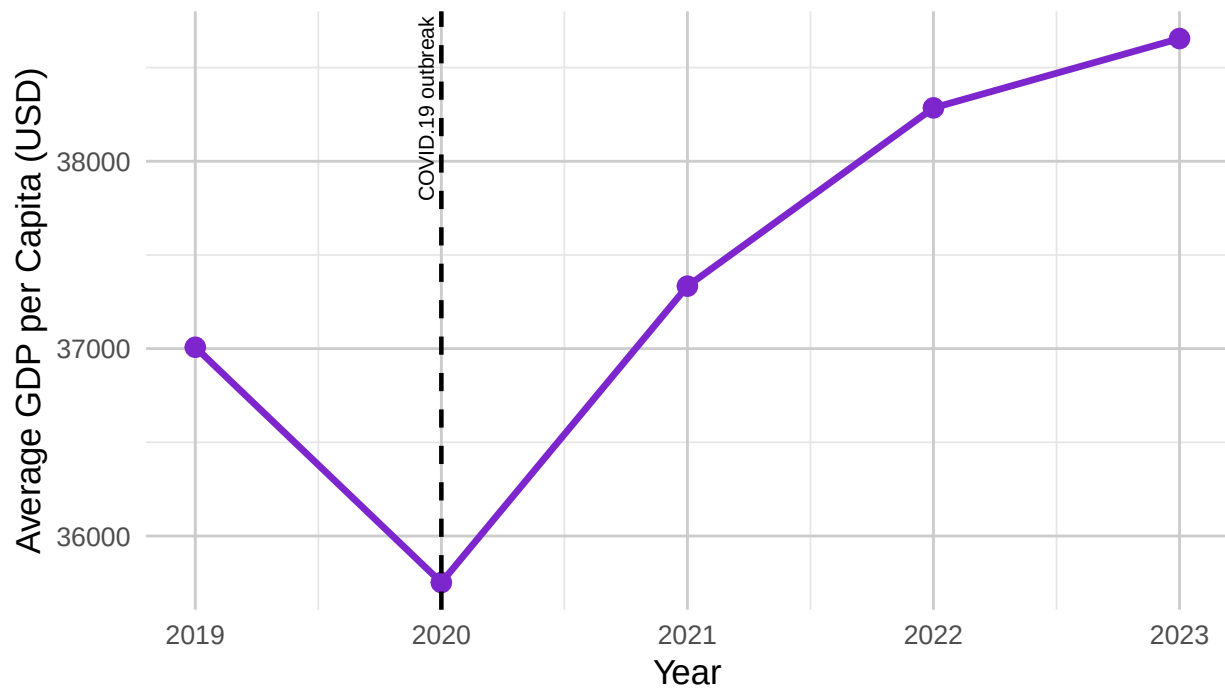


3.6 Event analysis

The initial COVID-19 outbreak occurred in December 2019. This slope graph indicates how life expectancy changed before and after the COVID-19 outbreak from 2019 and 2020 for 10 selected countries. The year 2019 shows pre-pandemic and 2020 shows post-outbreak. A surge in COVID-19 cases negatively impacted public health and GDP per capita worldwide during COVID-19 crisis and recovery time has been unequal across different countries. This graph demonstrates that while countries with higher GDP per capita maintained higher life expectancy in general, during the initial COVID-19 crisis, life expectancy was driven not only solely on GDP per capita, but also on population age, government support and healthcare structure.

Average GDP per Capita of 10 Countries (2019–2023)

Event analysis with COVID–19 outbreak highlighted



Source: World Bank indicators

Here you provide a description of why the plot above is relevant to your specific social problem.

Part 4 - Discussion

4.1 Discuss your findings

The results of this analysis show a clear relationship between GDP per capita and life expectancy across the ten selected countries between 2019 and 2023. Countries with higher GDP per capita generally demonstrated higher life expectancy, suggesting that stronger economies are associated with better healthcare systems, living conditions, and overall population health. The temporal analysis also showed a decline in average life expectancy between 2019 and 2021, followed by a recovery by 2023, which may reflect the global impact of the COVID-19 pandemic. In addition, the comparison between developed and developing countries highlighted persistent inequalities, indicating that economic development remains an important factor in explaining differences in health outcomes, although other external factors can also influence population well-being.

Part 5 - Reproducibility

5.1 Github repository link

Link to our PUBLIC repository here: <https://github.com/Brian921013/Group-6-assignment>

5.2 Reference list

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